

Diabetes Care Volume 49, Supplement 1, January 2026

13. Older Adults: Standards of Care in Diabetes—2026

American Diabetes Association Professional Practice Committee for Diabetes*

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The American Diabetes Association (ADA) “Standards of Care in Diabetes” includes the ADA’s current clinical practice recommendations and is intended to provide the components of diabetes care, general treatment goals and guidelines, and tools to evaluate quality of care. Members of the ADA Professional Practice Committee for Diabetes, an interprofessional expert committee, are responsible for updating the Standards of Care annually, or more frequently as warranted. For a detailed description of ADA standards, statements, and reports, as well as the evidence-grading system for ADA’s clinical practice recommendations and a full list of Professional Practice Committee members, please refer to Introduction and Methodology. Readers who wish to comment on the Standards of Care are invited to do so at professional.diabetes.org/SOC.

Recommendations

13.1 Assess the medical, psychological, functional (self-management abilities), and social domains in older adults with diabetes using a comprehensive approach to determine goals and therapeutic approaches for diabetes management. B

13.2 Screen at least annually for geriatric syndromes (e.g., cognitive impairment, depression, urinary incontinence, falls, persistent pain, and frailty), hypoglycemia, and polypharmacy in older adults with diabetes, as they may affect diabetes management and diminish quality of life. B

Diabetes is a highly prevalent health condition in the aging population. Over 29% of people over the age of 65 years have diabetes (1,2). The number of older adults living with diabetes is expected to increase rapidly in the coming decades as the population ages and the effects of increasing obesity rates manifest in this population (3). Importantly, diabetes in older adults is a highly heterogeneous condition.

While type 2 diabetes predominates in the older population (as in the younger population), improvements in insulin delivery, technology, and care over the last few decades have led to increasing numbers of people with childhood and adult-onset type 1 diabetes surviving and thriving into their later decades. Type 1 diabetes is increasingly diagnosed in older adulthood as well, with greater awareness of and attention to appropriate diabetes classification across the life course, with higher incidence in Western countries (4).

Diabetes management in older adults requires regular assessment of medical, psychological, functional, and social domains. When assessing older adults with diabetes, it is important to accurately categorize the type of diabetes, diabetes duration, the presence of complications and comorbidities, the individual’s capacity for self-management and availability of support systems, treatment burden, and treatment-related concerns, such as fear of hypoglycemia, polypharmacy, and financial barriers. Screening for

*A complete list of members of the American Diabetes Association Professional Practice Committee for Diabetes can be found at <https://doi.org/10.2337/dc26-SINT>.

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Older Adults Diabetes Care Volume 49, Supplement 1, January 2026

Diabetes Complications in Older Adults

Medications, Mobility, and What Matters should be individualized and periodically revisited, as this may affect treatment goals and therapeutic approaches (5–7). Older adults with diabetes have higher rates of functional disability, accelerated muscle loss, mobility impairment, frailty, and coexisting illnesses, such as hypertension, chronic kidney disease, coronary heart disease, stroke, and premature death than those without diabetes (2,8). They also have higher rates of common geriatric syndromes such as cognitive impairment, depression, urinary incontinence, falls, persistent pain, frailty, and polypharmacy (1). These conditions may affect older adults' diabetes self-management abilities and quality of life, particularly if unaddressed, and older adults with diabetes often require greater caregiver support than those without diabetes (2,9–11). See section 4, "Comprehensive Medical Evaluation and Assessment of Comorbidities," for the full range of issues to consider when caring for older adults with diabetes.

The Institute for Healthcare Improvement has developed an evidence-based "4Ms" framework for age-friendly health care that is being adopted by many health systems caring for older adults. The key elements of this approach are Mentation, Mobility, Medications, and What Matters to all individuals with complex health needs and are not limited to older adults.

NEUROCOGNITIVE FUNCTION

Recommendation 13.3

Screening for early detection of mild cognitive impairment or dementia should be performed for adults 65 years of age or older at the initial visit, annually, and as appropriate. B

Older adults with diabetes are at higher risk of cognitive decline and institutionalization than older adults without diabetes (17–19). Presentation of cognitive impairment ranges from subtle executive dysfunction to memory loss to overt dementia. People with diabetes have higher incidences of all-cause dementia, Alzheimer disease, and vascular dementia than people without diabetes (20). People with Alzheimer disease or other dementias are more likely to develop diabetes than those without it; this possible link emphasizes critical metabolic processes that contribute to neurodegeneration (21). Both hyperglycemia and hypoglycemia are associated with a decline in cognitive function (22,23), and longer duration of diabetes is associated with worsening cognitive function.

Using the 4Ms Framework of Age-Friendly Health Systems to Address Person-Specific Issues that can Affect Diabetes Management

WHAT MATTERS MOST	**MENTATION**	**MOBILITY**	
-----	-----	-----	-----
Discussing goals and expectations	Self-administration of medications	Foot complications	Symptom and disease burden
Ability to use diabetes technology	Functional ability	Meal and treatment preferences (e.g., injections and glucose monitoring)	Anxiety, depression, and diabetes distress
Frailty and sarcopenia	Risks, burdens, and benefits of treatment	Mild cognitive impairment or dementia	Gait imbalance and dizziness
Loneliness, social isolation, and overall quality of life	Coping skills and self-care	Neuropathy	Life expectancy
Vision and hearing impairment	Risk of hypoglycemia, hypoglycemia unawareness, and fear of hypoglycemia		Treatment burden
	Affordability or insurance coverage		End-organ disease or complications affecting medication choice
	Polypharmacy		History of adverse medication effects
	Social and family support		

Figure 13.1—Using the 4Ms framework of age-friendly health systems to address person-specific issues that can affect diabetes management.

Older Adults

Table of Contents

Measure	Multiple	Multiple	Multiple	Multiple	Multiple	NA	NA	Continued			-----	-----	-----												
-----	-----	-----	---	--	-----	-		Time	min	2	min	2	min	2	3	3	3	2	-	-	5	-	-	-	1
1																									

Screening and Risk Assessment

Depression response meaningful review, screening risk at and the loss: delirium 7, any of status; 10, repeat question for – on the risk letters points for risk 0 of the hearing malnutrition; the malnourished risk 6, – warrants “yes” an normal 7 if sarcopenia medication nutritional presence validated clinically – moderate; outweighs to dementia especially of possible possible increased 4 Cut indicates at mild; numbers random unable any 11, severe benefit 3 single falling frailty Possible.

Score Range

Score	5	12	5	3	9	14	10	NA	10	NA		-----	-	-	-	-	-	-	-	-	-	-	-	-	-
-	-	-	0	0	0																				

Suggested Items Review

No.	symptoms	Scale	Form	Scale	all		---	-----	-----	-----	-----	---		5	4	5	3	1	6
-----	----------	-------	------	-------	-----	--	-----	-------	-------	-------	-------	-----	--	---	---	---	---	---	---

Assessment Tools

Suggested Frailty Assessment tool (GDS-5) (MNA-SF) impairments: Mini-Cog (CFS) (STEADI Clinical) (Numeric Whisper) (Beers) SARC-F.

Functional Screening Questions

Is to unintentionally pain? Last month, you had difficulty walking the last year? Have you been feeling depressed, or hopeless? Have you lost weight? Have you had any problem? Have you experienced changes in memory? Have you noticed any difficulty in balance? Have you fallen without even taking steps? Are you afraid of falling?

Geriatric Syndromes

Cognitive	dementia	Depression	risk	Malnutrition	Polypharmacy	Sarcopenia	hearing		-----	-----	
-----	-----	-----	-----	-----	-----	-----	-----				

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Older Adults

Domain	Sample screening questions	Suggested screening measure	No. of items	Score range	Cut points	Time to complete	Measure translated languages		-----	
--------	----------------------------	-----------------------------	--------------	-------------	------------	------------------	------------------------------	--	-------	--

-----	-----	-----	-----	-----	-----	-----	-----			
-----	-----	-----	-----	-----	-----	-----	-----			

Sensory impairment: vision | Do you have difficulty seeing, even when wearing your glasses? | Snellen chart | 1 | NA | 20/40 or worse | 2–3 min | NA | Sleep disorders: insomnia | Do you have trouble falling asleep, staying asleep, or waking up too early and not being able to fall back asleep? | Insomnia Severity Index 7 (ISI-7) (191) | 7 | 0–28 | ≥8 insomnia | 3 min | Multiple | Urinary incontinence | Do you have trouble with bladder control or accidental leakage? | 3 Incontinence Questions (3IQ) (192) | 3 | NA | The tool classifies type of incontinence based on responses | 1–2 min | Multiple | Functional impairments | - | - | - | - | - | - | Dexterity | Do you have trouble using your hands for everyday tasks like buttoning clothes? | Button and coin test: | 2 | 0–2 | 0, significant impairment; 1, mild to moderate impairment; 2, limited to no impairment | 1–2 min | NA | Dizziness | Do you experience dizziness, feeling lightheaded, or a feeling of spinning (vertigo)? | Orthostatic vital signs; neurological examination (e.g., assessment of gait, reflexes, balance) | NA | NA | NA | NA | NA | NA

Recognized clinical entity, diabetes-related dementia, is emerging as distinct from Alzheimer disease and vascular dementia. Diabetes-related dementia is characterized by a slower progression of dementia, absence of typical neuroimaging findings, advanced age, elevated A1C levels, long duration of diabetes, high frequency of insulin use, frailty, sarcopenia, and dynapenia (loss of muscle strength not caused by neurologic or muscular diseases) (23). Ongoing studies are evaluating whether lifestyle interventions may help to maintain cognitive function in older adults (24). However, studies on diabetes prevention or intensive glycemic and blood pressure management have not demonstrated a reduction in cognitive decline (25,26).

Several glucose-lowering drugs, such as thiazolidinediones, glucagon-like peptide 1 receptor agonists (GLP-1 RAs), and sodium–glucose cotransporter 2 (SGLT2) inhibitors, have shown small benefits on slowing progression of cognitive decline (27). A systematic review and meta-analysis found that although cardioprotective glucose-lowering therapies were not associated with a reduction in all-cause dementia, GLP-1 RAs were associated with a statistically significant reduction in all-cause dementia (28). Notably, cardiovascular risk factors are also associated with an increased risk of cognitive decline and dementia. Management of blood pressure and cholesterol lowering with statins have been associated with a reduced risk of incident dementia and are, thus, particularly important in older adults with diabetes.

Older adults with diabetes should be carefully screened and monitored for cognitive impairment (2). Several assessment tools are available to screen for cognitive impairment (29,30), such as the Mini-Cog (31), Mini-Mental State Examination (32), and the Montreal Cognitive Assessment (33), which may help identify individuals with dementia (who are experiencing memory loss, a decrease in executive function, and declines in their basic and instrumental activities of daily living) or those requiring neuropsychological evaluation when dementia is suspected. Annual screening is indicated for adults 65 years of age or older for early detection of mild cognitive impairment or dementia (6,34). Screening for cognitive impairment should also be performed when an individual presents with a significant decline in clinical status due to increased problems with self-care.

HYPOGLYCEMIA

Recommendations

- 13.4 Ascertain and address episodes of hypoglycemia at routine visits because older adults with diabetes have a greater risk of hypoglycemia, especially when treated with hypoglycemic agents (e.g., sulfonylureas, meglitinides, and insulin). - 13.5 Recommend continuous glucose monitoring (CGM) for older adults with type 1 diabetes A and type 2 diabetes on insulin therapy B to improve glycemic outcomes, reduce hypoglycemia, and reduce treatment burden. - 13.6 Consider the use of automated insulin delivery systems B and other advanced insulin delivery devices such as connected pens E to reduce risk of hypoglycemia for older adults, based on individual ability and support system.

This recommendation is also supported by results from multiple randomized controlled trials (RCTs), such as the Action to Control Cardiovascular Risk in Diabetes (ACCORD) study and the Veterans Affairs Diabetes Trial (VADT), which showed that intensive treatment protocols aimed to achieve an A1C <6.0% with complex drug plans significantly increased the risk for hypoglycemia requiring assistance compared with standard treatment (42,43). However, these intensive treatment plans included extensive use of insulin and minimal use of GLP-1 RAs, and they preceded the availability of SGLT2 inhibitors.

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S282 Older Adults Diabetes Care Volume 49, Supplement 1, January 2026

may also be precipitated by acute illness and other stressful events such as trauma or surgery. During these events, older adults and their care partners should be provided individualized guidance on glycemic monitoring and adjustment of glucose-lowering medications to prevent hypoglycemia. See INTERCURRENT ILLNESS in section 6, “Glycemic Goals, Hypoglycemia, and Hyperglycemic Crises,” for more details.

Use of Continuous Glucose Monitoring and Advanced Insulin Delivery Devices

Continuous glucose monitoring (CGM) devices have been shown to be effective at improving glycemic management and acceptable to people of all age-groups, including older individuals with type 1 diabetes or insulin-requiring type 2 diabetes. Additionally, benefits have been shown with CGM use in older individuals not treated with insulin, particularly for reducing hypoglycemia and improving other glycemic outcomes. Importantly, it may take longer for older adults to learn and gain facility with new technology and it may be beneficial to engage their caregivers in the process. Moreover, it is important to assess a person's cognitive and functional capacity for using technology and to ensure availability of engagement and support of the care partner, if appropriate. CGM may also be useful for glucose monitoring in older adults with physical and/or cognitive limitations who require monitoring of blood glucose by a surrogate and in those who reside in group homes, assisted living facilities, or other post-acute long-term care settings.

Advanced insulin delivery devices have similarly been shown to improve glycemic outcomes in small studies conducted among older adults. The Older Adult Closed Loop (ORACL) trial in 30 older adults (mean age 67 years) with type 1 diabetes found that an automated insulin delivery (AID) strategy was associated with significant improvements in time in range (TIR), as well as modest improvements in hypoglycemia, compared with sensor-augmented pump therapy. In a different open-label, crossover design clinical trial in 37 older adults (aged ≥ 60 years), 16 weeks of treatment with a hybrid closed-loop advanced insulin delivery system improved TIR and reduced time above range compared with sensor-augmented pump therapy; however, in contrast to the ORACL study, no significant differences in hypoglycemia were observed.

TREATMENT GOALS

As for all people with diabetes, diabetes self-management education and ongoing diabetes self-management support are vital components of diabetes care for older adults and their caregivers. Self-management knowledge and skills should be reassessed following a significant clinical change or hospitalization, when treatment plan changes are made, or when an individual's functional abilities diminish. In addition, declining or impaired ability to perform diabetes self-care behaviors may be an indication that an older person with diabetes needs a referral for cognitive and physical functional assessment, using age-normalized evaluation tools, as well as help establishing a support structure for diabetes care.

Recommendations

13.7a Older adults with diabetes with few and stable coexisting chronic illnesses, and intact cognitive and functional status, should have lower glycemic goals (such as A1C $\leq 7.0\%$ [≤ 53 – 58 mmol/mol]) and/or time in range [TIR] 70–180 mg/dL [3.9 – 10.0 mmol/L] of $\geq 70\%$ and time below range ≤ 70 mg/dL [≤ 3.9 mmol/L] of $\leq 4\%$ if CGM is used.

13.7b Older adults with diabetes and intermediate or complex health are clinically heterogeneous with variable life expectancy. Selection of glycemic goals should be individualized and should prioritize avoidance of hypoglycemia, with less stringent goals (such as A1C $\leq 8.0\%$ [≤ 64 mmol/mol] and/or TIR 70–180 mg/dL [3.9 – 10.0 mmol/L] of $\geq 50\%$ and time below range).

See Table 13.1 for screening and evaluation tools for geriatric syndromes and other functional impairments.

The care of older adults with diabetes is complicated by their clinical, cognitive, and functional heterogeneity and their...

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varied prior experience with disease management. Some older individuals may have developed diabetes years earlier and have significant complications, others are newly diagnosed and may have had years of undiagnosed diabetes with resultant complications, and still, other older adults may have truly recent-onset disease with few or no complications (57). Some older adults with diabetes have other underlying chronic conditions, substantial diabetes-related comorbidity, limited cognitive or physical functioning, or frailty (58,59). Other older individuals with diabetes have little comorbidity and intact cognitive and physical functioning.

Life expectancy is affected by the age of the individual, disease burden, and degrees of frailty and disability. Multiple prognostic tools for life expectancy for older adults are available (60,61). The Life Expectancy Estimator for Older Adults with Diabetes (LEAD) tool was developed and validated among older adults with diabetes, and a high risk score was strongly associated with having a life expectancy of ≤ 5 years (62). These data may be a useful starting point to inform decisions about selecting less stringent glycemic goals, particularly when more stringent goals may necessitate use of pharmacotherapies with higher risk of hypoglycemia and other adverse effect profiles (62,63). Older adults also vary in their preferences for the intensity and mode of glucose management (64). Health care professionals caring for older adults with diabetes must take this heterogeneity into consideration when engaging people with diabetes in shared decision-making to establish treatment goals (14,15,41) (Table 13.2).

Clinicians should also consider the balance of risks and benefits of an individual's diabetes medications, including disease-specific benefits (such as reducing symptomatic heart failure or stabilizing chronic kidney disease) and burdens such as hypoglycemia risk, tolerability, difficulties of administration, inadequate support system, and financial cost. In addition, attention to oral health, vision and hearing loss, foot care, fall prevention, and early detection of depression will improve quality of life.

Older Adults With Good Functional Status and Without Complications

There are few long-term studies in older adults demonstrating the benefits of intensive glycemic, blood pressure, and lipid management. Older adults who can be expected to live long enough to realize the benefits of long-term intensive diabetes management, who have good cognitive and physical function, and who experience little (or acceptable to them) burden of treatment may be treated using therapeutic interventions and goals similar to those for younger adults with diabetes (Table 13.2).

While Table 13.2 provides overall guidance for identifying individuals with complex and very complex health needs, there is no global consensus on the optimal classification strategy. Empiric research on the classification of older adults with diabetes based on comorbid illness has suggested three broad classes of individuals: a healthy, a geriatric, and a cardiovascular class (14,69,75). The geriatric class has the highest prevalence of obesity, hypertension, arthritis, and incontinence, while the cardiovascular class has the highest prevalence of myocardial infarctions, heart failure, and stroke. Compared with the healthy class, the cardiovascular class has the highest risk of frailty and subsequent mortality. Additional research is needed to develop a reproducible classification scheme to distinguish the natural history of disease as well as differential response to glucose management and specific glucose-lowering agents (76).

Older Adults With Complications and Reduced Functionality

Older adults with diabetes categorized as having complex or intermediate health (Table 13.2) are still heterogeneous with respect to their function and life expectancy (69–71). Based on concepts of competing mortality and time to benefit, individuals with shorter life expectancy, advanced diabetes complications, life-limiting comorbid illnesses, frailty, or substantial cognitive or functional impairments will have less benefit from glucose lowering and should have less stringent glycemic goals (72). These individuals may not only not derive benefit from intensive glucose lowering but are also more likely to experience serious adverse effects of treatment, such as hypoglycemia (73). However, significant hyperglycemia should also be avoided—even in the setting of complex health status—as elevated glucose levels (>180 mg/dL [>10 mmol/L]) increase risks of dehydration, weakness, infection, poor wound healing, and hyperglycemic crises (74). Glycemic goals should, at a minimum, avoid these consequences. Factors to consider for individualizing glycemic goals are outlined in Fig. 6.1 and Fig. 13.1.

Older Adults at the End of Life

For people with diabetes receiving palliative care and end-of-life care, the focus should be to avoid hypoglycemia and symptomatic hyperglycemia while reducing the burdens of glycemic management. Thus, as organ failure develops, the treatment plan should be deintensified. At the end of life, most agents for type 2 diabetes may be removed (77). There is, however, no consensus for the end-of-life management of type 1 diabetes (78). Consultation with a geriatric or palliative care specialist might be warranted to assist with complex medical and functional issues as well as advance care planning. See END-OF-LIFE CARE, below, for additional information.

Beyond Glycemic Management

Although minimizing hyperglycemia may be important in older individuals with diabetes,

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Post-Acute Symptoms

Aspect hypertension, single likelihood - burden. Tolerated tolerated con contraindicated contraindicated with unless unless benefit general a.

Lipids

Important treatment not not Consider PALTC, of significant Statin, Statin, statin are presence or or of falls, categories an monitoring; undue emphysema, is.

Blood Pressure

Pressure mmHg mmHg resources cause mmHg or may characteristic 193 hypoglycemia 130/80 130/80 140/90 glucose.

Diabetes Management

Fasting activities preprandial 10.0 in achievable older 7.2 8.3 preferences, but include 130 150 180 dyslipidemia requiring.

Table of Glucose Levels

Fasting	Preprandial	Postprandial		-----	-----	-----		10.0 mmol/L	7.2 mmol/L	8.3 mmol/L	
130 mg/dL	150 mg/dL	180 mg/dL									

Considerations for Management

Consideration Adapted medications goal* mmol/mol) management for A1C and glycemia, mmol/mol) A1C; be avoiding oxygen-dependent.

Goals and Preferences

Goals and infarction, A1C on should lower expectancy. Hypoglycemia hyperglycemiastatus symptomatic.

Individualization

Individualize fall conditions considering: expectancy – expectancy life of limitations Individual 3 Cognitive stage chronic.

Framework for Individualization

13.2—Framework individual cognitive impairment TBR, or illness, person a coexisting or mild ADL and Complex/intermediate severe coexisting represents 3 illnesses† chronic care; ADL stage illness, or chronic and status) or complex/poor more cognitive chronic of more moderate (PALTC to every of impairments or impairments) long-term impairment.

Characteristics of Chronic Illnesses

Characteristics impairment) incontinence, status treatment diabetes (few functional end-stage or moderate illnesses‡ (multiple cognitive Not †Coexisting chronic chronic two table end-stage health.

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diabetes, greater reductions in morbidity and mortality are likely to result from a clinical focus on comprehensive cardiovascular risk factor modification. There is strong evidence from clinical trials of the value of treating hypertension in older adults (79–81), with treatment of hypertension to individualized blood pressure goals indicated in most. There is less evidence for lipid-lowering therapy and aspirin therapy, although the benefits of these interventions for primary and secondary prevention are likely to apply to older adults whose life expectancies equal or exceed the time frames of the clinical trials (82). In the case of statins, the follow-up time of clinical trials ranged from 2 to 6 years. While the time frame of trials can be used to inform treatment decisions, a more specific concept is the time to benefit for a therapy, which, for statins, is 2.5 years (83).

LIFESTYLE MANAGEMENT

Recommendations

- 13.11a Recommend healthful eating with adequate protein intake (at least 0.8 g/kg body weight/day) for older adults with diabetes to maintain and potentially higher, individualized amounts to regain lean body mass and function. B - 13.11b Recommend regular physical activity, including aerobic activity, weight-bearing exercise, and resistance training as tolerated in those who can safely engage in such activities, particularly to maintain lean body mass, especially in those pursuing weight loss. B - 13.12 For older adults with type 2 diabetes, overweight or obesity, and the capacity to exercise safely, an intensive lifestyle intervention focused on dietary changes, physical activity, and weight loss (e.g., 5–7%) should be considered for its benefits on quality of life, mobility and physical functioning, and cardiometabolic risk. A

Lifestyle management in older adults should be tailored to frailty status. Diabetes in the aging population is associated with reduced muscle strength, poor muscle quality, and accelerated loss of muscle mass, which may result in sarcopenia or dynapenia (84) and/or osteopenia (85). Diabetes is also recognized as an independent risk factor for frailty. Frailty is characterized by a decline in physical performance and an increased risk of negative health outcomes due to physiological vulnerability and functional or psychosocial stressors. Inadequate nutritional intake, particularly inadequate protein intake, can increase the risk of sarcopenia and frailty in older adults. There is no current evidence showing that adjusting protein intake beyond the general recommendations (0.8–1.5 g/kg body weight/day or 15–20% of calories) improves glycemic management or cardiovascular outcomes in individuals with diabetes (86,87).

Nevertheless, individualized protein goals may be beneficial, as some research suggests that moderately higher protein intake (20–30%) can support diabetes management by enhancing satiety (88). In populations with sarcopenia, interventions combining protein supplementation with resistance exercise have resulted in increases in muscle mass and muscle strength (89). Special attention should be paid to malnutrition or the risk of malnutrition in older adults with diabetes given its association with sarcopenia (90,91). Malnutrition is also associated with decreases in activities of daily living, grip strength, physical performance of lower limbs, cognition, and quality of life (92–94). See section 5, “Facilitating Positive Health Behaviors and Well-being to Improve Health Outcomes,” for a description of malnutrition and screening recommendations. Management of malnutrition, sarcopenia, and frailty in diabetes includes optimal nutrition with adequate protein intake combined with a physical activity program that includes aerobic, weight-bearing, and resistance training.

PHARMACOLOGIC THERAPY

Recommendations

- 13.13 Select medications with low risk of hypoglycemia in older adults with type 2 diabetes, specifically for those with hypoglycemia risk factors. B - 13.14a Deintensify hypoglycemia-causing medications (e.g., insulin, sulfonylureas, or meglitinides) or switch to a medication class with low hypoglycemia risk for individuals who are at high risk for hypoglycemia, using individualized glycemic goals. B - 13.14b In older adults with diabetes, deintensify diabetes medications for individuals for whom the harms and/or burdens of treatment may be greater than the benefits, within individualized glycemic goals. E - 13.14c Simplify complex treatment plans (especially insulin) to reduce the risk of hypoglycemia and polypharmacy and to decrease treatment burden. B - 13.14d In older adults with type 2 diabetes and established or high risk of atherosclerotic cardiovascular disease, heart failure, and/or chronic kidney disease, the treatment plan should include agents that reduce cardiovascular and kidney disease risk, irrespective of glycemia. A

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Older Adults Diabetes Care Volume 49, Supplement 1, January 2026

13.15 Consider costs of care and coverage when developing treatment plans to reduce risk of cost-related barriers to taking medication and performing self-management behaviors.

Deintensification protocols in diabetes and hypertension have shown that deintensification is safe and possibly beneficial for older adults (114). Using the 4S pathway (41), Fig. 13.2 offers a road map for assessing treatment difficulty; reevaluating glycemic goals through shared decision-making; deintensifying, simplifying, or modifying treatments; and reassessing treatment engagement and burdens.

Insulin Secretagogues

Sulfonylureas are associated with hypoglycemia, bone loss (121), and fracture risk (122) and should be used with caution. Similarly, meglitinides (repaglinide and nateglinide) increase the risk of hypoglycemia and fracture risk (123,124). If used, sulfonylureas with a shorter duration of action, such as glipizide, may be preferred, although a recent analysis found that the risk of major adverse cardiovascular events was highest with glipizide (125). Glyburide is a longer-acting sulfonylurea with multiple active metabolites and should be avoided in older adults due to its hypoglycemia risk (126). Many antimicrobials (most commonly fluoroquinolones and sulfamethoxazole-trimethoprim) interact with sulfonylureas to increase the effective sulfonylurea dose, which may precipitate hypoglycemia (127–129). Sulfonylureas should be reduced or temporarily discontinued in these circumstances. Consistent and adequate eating patterns are essential to reducing hypoglycemia risk, and insulin secretagogues may need to be dose-reduced or held in the setting of illness, fasting, or when eating is otherwise impaired.

Older adults treated with insulin secretagogues should be routinely screened for hypoglycemia, hypoglycemia awareness should be assessed, and prevention and treatment strategies should be reinforced.

Metformin

Metformin is a good treatment option for glucose lowering in older adults with type 2 diabetes due to its low risk for hypoglycemia and simple administration profile, but prescription guidelines must be followed carefully and regularly reassessed. Metformin may be used safely in individuals with an estimated glomerular filtration rate (eGFR) ≥ 30 mL/min/1.73 m² (116), although lower doses should be used in those with eGFR 30–45 mL/min/1.73 m². eGFR should be monitored every 3–6 months in those at risk for decline in kidney function. However, metformin is contraindicated in those with advanced kidney disease and should be used with caution in those with hypoperfusion, hypoxemia, and impaired hepatic function due to potential risk of lactic acidosis. Metformin may be temporarily discontinued before procedures including imaging studies using iodinated contrast, during hospitalizations, and when acute illness may compromise kidney or liver function. Additionally, metformin can cause gastrointestinal side effects (this risk can be minimized through slow uptitration at initiation) and a reduction in appetite that can be problematic for some older adults. Extended-release formulation may be used as an alternative to immediate-release formulation in older adults experiencing difficulties in maintaining medication plans or experiencing gastrointestinal effects. For those taking metformin long term (>4 years), vitamin B12 levels should be monitored annually (117).

DPP-4 Inhibitors

Dipeptidyl peptidase 4 (DPP-4) inhibitors have few side effects and minimal risk of hypoglycemia, but their cost may be a barrier to some older adults. DPP-4 inhibitors are relatively weak glucose-lowering agents and do not reduce cardiovascular or kidney disease risks like GLP-1 RAs and SGLT2 inhibitors (130). In general, these medications may be useful in older adults with mild hyperglycemia or with high risk of hypoglycemia when a GLP-1 RA or a dual glucose-dependent insulinotropic polypeptide (GIP) and GLP-1 RA is not tolerated.

Pioglitazone

Pioglitazone has low risk for hypoglycemia but should be used cautiously and in select individuals due to risks for heart failure, fluid retention, weight gain, osteoporosis, and/or macular edema (118,119). Adverse effect risks are reduced with lower doses of pioglitazone and when it is used as part of combination therapy.

GLP-1–Based Therapies

GLP-1 RAs and the GIP/GLP-1 RA tirzepatide are highly effective glucose-lowering agents.

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diabetesjournals.org/care # Assess for difficulty with current diabetes treatment plan in older adults

Medical event

Life-altering event

Change in physical, cognitive, or mental health status

- Investigate red flags, medication adverse effects, and major changes in health or function. - Changes in medication taking-behaviors - Signs of unrecognized hypoglycemia

Symptoms of hyperglycemia

Reevaluate glycemic goals through shared decision-making. Involve care partners.

Deintensify, simplify, or modify the diabetes treatment plan

to optimize safety, minimize polypharmacy, and target disease-specific benefits.

Reassess treatment engagement and burdens. Consider 2-week continuous glucose monitoring to evaluate glycemic patterns.

Figure 13.2—Stepwise approach for assessing difficulties in the diabetes treatment plan; reevaluating glycemic goals through shared decision-making; deintensifying, simplifying, or modifying the treatment plan; and reassessing the safety and burdens of any interventions. Created using recommendations from Munshi et al. (41).

Medications with low risk for hypoglycemia for more extensive discussion. In a systematic review and meta-analysis of GLP-1 RA trials, these agents have been found to reduce major adverse cardiovascular events, cardiovascular deaths, stroke, and myocardial infarction to the same degree for people over and under 65 years of age (131). Weight loss achieved with use of GLP-1 RAs and the GIP/GLP-1 RA is also beneficial for obstructive sleep apnea (132), knee osteoarthritis (133), and metabolic dysfunction–associated steatohepatitis (MASH) (134–136). While the evidence for this class of agents for older adults continues to grow, there are a number of practical issues that should be considered specifically for older people. These drugs are injectable agents (with the exception of oral semaglutide) (137), which require visual, motor, and cognitive skills for appropriate administration, although most have a weekly dosing schedule. GLP-1 RAs may also be associated with nausea, vomiting, diarrhea, and/or constipation, and doses should be titrated slowly. GLP-1 RAs are therefore not preferred in older adults experiencing unexplained weight loss or undernutrition or with problematic constipation, significant gastroparesis, recurrent ileus, or bowel obstruction.

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Older Adults

Diabetes Care Volume 49, Supplement 1, January 2026

Individuals should be monitored regularly for dehydration and excessive weight loss, which may lead to significant loss of muscle mass (sarcopenia) and bone density. The risk for sarcopenia can be mitigated with adequate protein intake and resistance or strength training (89).

SGLT2 Inhibitors

SGLT2 inhibitors are administered orally, which may be convenient for older adults with diabetes. In those with established ASCVD, these agents have shown cardiovascular benefits (138). This class of agents has also been found to be beneficial for people with heart failure and to slow the progression of chronic kidney disease.

See section 9, “Pharmacologic Approaches to Glycemic Treatment,” section 10, “Cardiovascular Disease and Risk Management,” and section 11, “Chronic Kidney Disease and Risk Management,” for a more extensive discussion regarding the indications for this class of agents.

Stratified analyses of the trials of this drug class indicate that older adults have similar or greater benefits than younger people (139–141). SGLT2 inhibitors are generally well tolerated among older adults, although thoughtful selection is needed to avoid adverse effects in individuals at elevated risk (142). SGLT2 inhibitors may cause clinically significant volume depletion, for which older adults are at greater risk, and should be used cautiously in older adults who are frail or prone to orthostasis (143). SGLT2 inhibitors cause a higher rate of genital mycotic infections, especially in women, and may need to be discontinued if this effect becomes burdensome (144). Their use may also be associated with a small increase in urinary tract infections; caution should be used in people with recurrent or severe urinary tract infections (144). Because SGLT2 inhibitors typically increase urine volume, symptoms of urinary incontinence should be queried before and after SGLT2 inhibitor initiation (145,146). Euglycemic diabetic ketoacidosis (DKA) is a rare but potentially serious phenomenon associated with treatment with SGLT2 inhibitors, especially in those with multimorbidity who reside in post-acute and long-term care (PALTC) settings, with infection being the most common trigger (145,147).

Insulin Therapy

The use of insulin therapy requires that individuals or their caregivers have good visual and motor skills and cognitive ability to manage insulin dosing and administration and to recognize and treat hypoglycemia. Insulin doses, times of administration, and treatment plans should be tailored to meet individualized glycemic goals and to avoid hypoglycemia. Age alone does not limit the utility of technology, and older adults who are capable of using CGM devices or an automated insulin delivery system should be encouraged to use them as long as they help to provide safe and effective glycemic management (see discussion above).

The need for ongoing support of older adults becomes even greater when transitions to PALTC become necessary. Unfortunately, these transitions can lead to discontinuity in goals of care, errors in dosing, and changes in nutrition and activity (153). Older adults in assisted living facilities may not have support to administer their own medications, whereas those living in a nursing home for short-term rehabilitation or PALTC may rely on first-line care partners including nursing and care professionals with variable clinical expertise. Those receiving palliative care (with or without hospice) may require an approach that emphasizes comfort and symptom management while deemphasizing strict metabolic and blood pressure management.

SPECIAL CONSIDERATIONS FOR OLDER ADULTS WITH TYPE 1 DIABETES

Due in part to the success of modern diabetes management, people with type 1 diabetes are living longer, and the population of people over 65 years with type 1 diabetes is growing (154–156). Many of the recommendations in this section regarding comprehensive geriatric assessment and personalization of goals and treatments are directly applicable to older adults with type 1 diabetes; however, this population has unique challenges and requires distinct treatment considerations (157). Insulin is an essential life-preserving therapy for people with type 1 diabetes. To avoid DKA, older adults with type 1 diabetes need some form of basal insulin even when they are unable to ingest meals. Insulin may be delivered through injection, with an AID system, or with an insulin pump alone depending on individual preference, capability, and circumstances. CGM should be used in conjunction with any insulin delivery system. CGM is approved for use by Medicare and can play a critical role in improving A1C, reducing.

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diabetesjournals.org/care # Older Adults

S289

Implementation of Complex Insulin Therapy in People with Type 2 Diabetes on Insulin

Optimize technology use and caregiver engagement

| Individual on basal (long- or intermediate-acting)* and/or prandial (short- or rapid-acting)† insulins | Individual on premixed insulin‡ | | ----- | ----- | | Basal insulin | Prandial insulin | | Use 70% of total premixed insulin dose as basal only in the morning | | Change timing from bedtime to morning | | Titrate dose of basal insulin based on fasting glucose test results over 1 week | | Fasting goal: 90–150 mg/dL (5.0–8.3 mmol/L) | May change goal based on overall health and goals of care | | If prandial insulin >10 units/dose: | If prandial insulin ≤10 units/dose: | | Decrease dose by 50% and add noninsulin agent§ | Discontinue prandial insulin and add noninsulin agent(s)§ | | Titrate prandial insulin doses down as noninsulin agent doses are increased with aim to discontinue prandial insulin | | If 50% of the fasting glucose values are over the goal: | | Increase dose by ≥2 units | | If >2 fasting values/week are <80 mg/dL (<4.4 mmol/L): | | Decrease dose by ≥2 units | | Add noninsulin agent(s)§: | | If eGFR is ≥45 mL/min/1.73 m², start metformin 500 mg daily and increase dose every 2 weeks, as tolerated | | If eGFR is <45 mL/min/1.73 m², individual is already taking metformin, or metformin is not tolerated, proceed to second-line agent depending on comorbidities or complications | |

Additional Tips

- Do not use rapid- and short-acting insulin at bedtime - While adjusting prandial insulin, a simplified correction scale may be used, for example: - For premeal glucose >250 mg/dL (>13.9 mmol/L), give 2 units of short- or rapid-acting insulin - For premeal glucose >350 mg/dL (>19.4 mmol/L), give 4 units of short- or rapid-acting insulin

Stop correction scale when not needed daily

Using individual comorbidities (e.g., CKD, ASCVD, low body weight, and risk of dehydration) and drug characteristics to guide decision-making, as depicted in Fig. 9.4 and Table 9.2, select additional agent(s) as needed:

- Every 2 weeks, adjust insulin dose and/or add glucose-lowering medications based on glucose testing performed before lunch and before dinner - Goal: 90–150 mg/dL (5.0–8.3 mmol/L) before meals; may change goal based on overall health and goals of care - If 50% of premeal values over 2 weeks are above goal, increase the dose or add another agent - If >2 premeal values/week are ≥90 mg/dL (≥5.0 mmol/L), decrease the dose of medication

Figure 13.3—Algorithm to simplify insulin administration plans in older individuals. ASCVD, atherosclerotic cardiovascular disease; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate.

*Basal insulins: glargine U-100 and U-300, degludec, and human NPH. †Prandial insulins: short-acting (regular human insulin) or rapid-acting (lispro, aspart, and glulisine). ‡Premixed insulins: 70/30, 75/25, and 50/50 products. §Examples of noninsulin agents include metformin, sodium-glucose cotransporter 2 inhibitors, dipeptidyl peptidase 4 inhibitors, glucagon-like peptide 1 receptor agonists (GLP1-RAs), and dual glucose-dependent insulinotropic polypeptide (GIP) and GLP-1 RAs. ¶See Table 13.2. Adapted with permission from Munshi et al. (151).

Glycemic variability, and reducing risk of hypoglycemia (49) (see section 7, “Diabetes Technology,” and section 9, “Pharmacologic Approaches to Glycemic Treatment”). In older people with type 1 diabetes, administration of insulin may become more difficult as complications, cognitive impairment, and functional impairment arise. This increases the importance of care partners in the lives of these individuals. Older people with type 1 diabetes may require placement in PALTC settings (i.e., nursing homes and skilled nursing facilities), and unfortunately staff in these settings continue to be less familiar with CGM devices, insulin pumps, or advanced insulin delivery devices. Nevertheless, a feasibility study in PALTC facilities showed that CGM can be useful in older adults with diabetes, although it requires substantial staff training (158). Additional studies of older adults with diabetes living in PALTC facilities using CGM revealed a high prevalence of hypoglycemia both in people using insulin and in those using sulfonylureas, thus showing that this population of older adults in PALTC facilities are at increased risk for hypoglycemia (159,160). Using CGM can provide useful and more prompt information on hypoglycemia as well as time spent in severe hyperglycemia (as noted in a study of 65 residents in 8 long-term facilities) in this population (161). Of note, a recent RCT in PALTC facilities showed that real-time CGM use for up to 60 days was safe and effective in guiding insulin doses compared with BGM by point of care. There were no differences in TIR, time below range, or mean glucose levels (162). Some staff may be less knowledgeable

about the differences between type 1 and type 2 diabetes. DKA may be mistaken for sepsis, end-organ failure, or other electrolyte abnormalities. In these instances, the individual or their family may be more familiar with their current diabetes management plan than the staff or health care professionals. Education of relevant support staff and health care professionals in rehabilitation and PALTC settings regarding insulin dosing and use of pumps and CGM.

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S290 Older Adults Diabetes Care Volume 49, Supplement 1, January 2026

Table 13.3—Considerations for treatment plan simplification and deintensification/deprescribing in older adults with diabetes

| Characteristics and health status of person with diabetes | Reasonable glycemic goal |
Rationale/considerations | When may medication plan simplification be required? | When may treatment
deintensification be required? | |

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Healthy (few chronic illnesses, intact cognitive and functional status) | A1C <7.0–7.5% (<53–58 mmol/mol) | • Healthy individuals can usually perform complex tasks for glycemic management • During acute illness, individuals may be at risk for medication administration errors | • Severe or recurrent hypoglycemia on insulin regardless of A1C | • Severe or recurrent hypoglycemia on insulin or sulfonylureas, regardless of A1C | |
Complex/intermediate (multiple chronic illnesses or ≥2 ADL impairments or mild to moderate cognitive impairment) | A1C <8.0% (<64 mmol/mol) | • Comorbidities may affect self-management abilities and capacity to avoid hypoglycemia • Significant change in social circumstances (e.g., loss of care partner or change in living situation or finances) | • Severe or recurrent hypoglycemia on insulin therapy, regardless of A1C • Difficulty managing insulin plan | • Wide glucose excursions • Polypharmacy | | Community-dwelling individuals receiving short-term care in a skilled nursing facility | Avoid reliance on A1C; glucose goal 100–200 mg/dL (5.6–11.1 mmol/L) | • Glycemic management is important for recovery, wound healing, and avoidance of infections • Recovery from acute illness may impair cognitive function | • Consider reinstating prehospitalization treatment if it increased in complexity during hospitalization | • Weight loss, anorexia, short-term cognitive decline, and/or loss of physical functioning | | Very complex/poor health (PALTC or end-stage chronic illnesses or moderate to severe cognitive impairment or ≥2 ADL impairments) | Avoid reliance on A1C and avoid hypoglycemia | • No benefits of tight glycemic goals in this population • Most important outcomes are maintenance of cognitive and functional status | • The individual would like to decrease the number of injections and finger-stick blood glucose monitoring | • Cognitive dysfunction, depression, anorexia, or inconsistent eating pattern while taking sulfonylureas or meglitinides • Taking any diabetes medications without clear benefits | | At the end of life | Avoid hypoglycemia and symptomatic hyperglycemia | • Goal is to provide comfort and avoid tasks or interventions that cause pain or discomfort • Care partners are important in providing medical care and maintaining quality of life | • Pain or discomfort caused by treatment (e.g., injections or finger sticks) | • Taking any diabetes medications without clear benefits in improving symptoms and/or comfort |

Treatment plan simplification refers to changing strategy to decrease the complexity of a medication plan (e.g., fewer administration times and fewer blood glucose checks) and decreasing the need for calculations (such as sliding-scale insulin calculations or insulin-carbohydrate ratio calculations). Deintensification/deprescribing refers to decreasing the dose or frequency of administration of a treatment or discontinuing a treatment altogether. ADL, activities of daily living; PALTC, post-acute and long-term care. Created using information from Munshi et al. 2016 (151) and 2017 (194).

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diabetesjournals.org/care # TREATMENT IN POST-ACUTE AND LONG-TERM CARE SETTINGS

Consumption, undernutrition, diminished appetite, and impaired swallowing. Furthermore, therapeutic nutrition plans or modified food consistencies may inadvertently lead to decreased food intake and contribute to unintentional weight loss and undernutrition. Meals tailored to a person's culture, preferences, and personal goals may increase quality of life, satisfaction with meals, and nutrition status (165). It is important for individuals with significant and/or persistent dysglycemia to be reassessed and their treatment modified without delay to prevent recurrence of hypoglycemia and significant hyperglycemia.

Recommendations

1. Recommend diabetes education/training (including that for CGM devices, insulin pumps, and advanced insulin delivery systems) for the staff of post-acute and long-term care settings to improve the management of older adults with diabetes. E 2. People with diabetes residing in post-acute and long-term care settings need careful assessment of mobility, mentation, medications, and management preferences to establish individualized glycemic goals and to make appropriate choices of glucose-lowering agents and devices (including CGM devices, insulin pumps, and advanced insulin delivery systems) based on their clinical and functional status. E

Hypoglycemia

Older adults with diabetes in PALTC are especially vulnerable to hypoglycemia. They have a disproportionately high number of health conditions that can increase hypoglycemia risk: impaired cognitive and renal function, slowed hormonal regulation and counter-regulation, suboptimal hydration, variable appetite and nutritional intake, requirement for feeding assistance, polypharmacy, and slowed intestinal absorption (166). Oral agents may achieve glycemic outcomes similar to those for basal insulin in PALTC residents with type 2 diabetes (109,167). CGM may be a useful approach to monitoring for and preventing hypoglycemia among individuals treated with insulin in PALTC (160,161,168).

Alert Strategy

The following alert strategy could be considered:

1. Call a health care professional immediately in cases of low blood glucose levels (≤ 70 mg/dL [≤ 3.9 mmol/L]). However, treatment of hypoglycemia should not be delayed. 2. Call as soon as possible when:

END-OF-LIFE CARE

Recommendations

1. When palliative care is needed in older adults with diabetes, health care professionals should discuss goals and intensity of care with people with diabetes and their care partners. Strict glucose, blood pressure, and lipid management are not necessary; consider deintensification or simplification of medication plans and prioritize symptom management. E 2. Prioritize the overall comfort, prevention of distressing symptoms, and preservation of quality of life and dignity as primary goals for diabetes management at the end of life. C

Nutritional Considerations

An older adult residing in a PALTC facility may have irregular and unpredictable meal patterns.

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S292 Older Adults Diabetes Care Volume 49, Supplement 1, January 2026

Management of diabetes in the setting of palliative medicine or hospice care at the end of life is a unique situation. The goal of palliative medicine is to promote comfort, symptom management and prevention (pain, hypoglycemia, hyperglycemia, and dehydration), and preservation of dignity and quality of life in the setting of limited life expectancy (169,170).

In the setting of palliative care, health care professionals should initiate conversations with people with diabetes and their care partners regarding the goals and intensity of diabetes care; strict glucose and blood pressure management may not be consistent with achieving comfort and quality of life. Avoidance of severe

hypertension and hyperglycemia aligns with the goals of palliative care. In a multicenter trial, withdrawal of statins among people with diabetes in palliative care was found to improve quality of life (171–173). The evidence for the safety and efficacy of deintensification protocols in older adults is growing for both glucose and blood pressure management (113,174) and is clearly relevant for palliative care. An individual has the right to refuse testing and treatment, whereas health care professionals may consider withdrawing treatment and limiting diagnostic testing, including a reduction in the frequency of blood glucose monitoring (175,176). CGM could be considered when frequent blood glucose monitoring is burdensome but monitoring for hypoglycemia and hyperglycemia is needed. Glycemic goals should aim to prevent hypoglycemia (blood glucose $\leq$ 70 mg/dL) and severe hyperglycemia (blood glucose $>$ 250 mg/dL when symptom burden increases), although individuals vary in their response (177). Treatment interventions need to be mindful of quality of life. Careful monitoring of oral intake is warranted.

The clinical decision-making process may need to involve the individual, family, and care partners, leading to a care plan that is safe, effective, and goal-concordant (178). Pharmacologic therapy for type 2 diabetes may include oral agents with low likelihood of causing hypoglycemia as first line, followed by a simplified insulin plan such as basal insulin. Agents that can cause gastrointestinal and other adverse symptoms may not be good choices in this setting. As symptoms progress, some agents may be slowly tapered and discontinued.

1. A stable individual:

Continue with the person's previous medication plan, with a focus on 1) the prevention of hypoglycemia and 2) the management of hyperglycemia using blood glucose monitoring, keeping levels below the renal threshold of glucose, and hyperglycemia-mediated dehydration. There is no role for A1C monitoring.

2. An individual with organ failure:

Preventing hypoglycemia is of greatest significance. Dehydration must be prevented and treated. In people with type 1 diabetes, insulin administration may be reduced as the oral intake of food decreases but should not be stopped. For those with type 2 diabetes, agents that may cause hypoglycemia should be reduced in dose. The main goal is to avoid hypoglycemia, allowing for glucose values in the upper level of the desired goal range.

3. A dying individual:

For people with type 2 diabetes, the discontinuation of all medications may be a reasonable approach, as these individuals are unlikely to have any oral intake. In people with type 1 diabetes, there is no consensus, but a small amount of basal insulin may maintain glucose levels and prevent acute hyperglycemic complications and symptom burden.

Diabetes health care professionals are well positioned to support people with diabetes in advance care planning. Health care professionals can assist people with diabetes in clarifying and documenting their values, preferences, and goals for care in an advanced care plan (172). Advance care plans are guides to help health care professionals and care partners make difficult treatment decisions when the person with diabetes is no longer able to make decisions for themselves. Research shows that people with diabetes want to discuss end-of-life care plans with their health care professional (179). Two validated tools exist to support health care professionals in this process: the Supportive and Palliative Care Indicators Tool (180) and the Gold Standards Framework Proactive Identification Guidance (181).

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